

Задача 1.  $P(A) \cdot P(B) = 0,12$

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) \\ = P(A) + P(B) - (1 - P(\overline{A \cup B}))$$

$$= 0,7 - 0,58 = 0,12 \Rightarrow$$

$$P(A)P(B) = P(A \cap B) \Rightarrow \text{га, независ. со.}$$

2.  $P(3 \text{ вера}) = \binom{5}{3} \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^2 \approx 8,79\%$

$$P(1, 3, 5 | \text{отвора на 3}) = \frac{1}{\binom{5}{3}} = \frac{1}{10}$$

3. Ако са независ, вариацията е 0.

Водим случај разгледаме по формула:  
(или по def):

$$\text{Cov}(X+Y, Z) = E[(X+Y)Z] - E(X+Y)E(Z)$$

$$\stackrel{\text{линеарност}}{=} E[XZ] - E(X)E(Z) \\ + E[YZ] - E(Y)E(Z) \\ = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$$

4. Нека  $X_i = \begin{cases} 1, & i\text{-то изпитание е успешно} \\ 0, & \text{иначе} \end{cases}, X_i \stackrel{\text{i.i.d.}}{\sim} \text{Ber}(\frac{1}{2})$

$$\text{Търсим } \text{Cov}(X_1 + \dots + X_7, (1-X_4) + \dots + (1-X_{10}))$$

$$\stackrel{\text{от 3.}}{=} 4 \cdot \text{Cov}(X_1, -X_1) = -4DX_1 = -1 \quad (\text{т.к. } \text{Cov}(X_i, X_j) = 0 \text{ за } i \neq j)$$

$\Rightarrow$  Търсената корелация е

$$\text{Cor}(\underbrace{X_1 + X_2}_Y, \underbrace{7 \cdot \sum_{i=1}^{10} X_i}_Z) = \frac{-1}{\sqrt{10} \sqrt{2}} = -\frac{1}{7 \cdot \frac{1}{4}} = -\frac{4}{7}.$$

5.  $P(\text{точно две 3 рунга}) =$

$P(\text{има две ми 16-а и 20-та карти  
и две две 21-ва и 30-та})$

$$= P(\text{има две ми 16-а и 20-та}) - P(\text{има две 19-а и 30-та})$$

$$= \frac{\binom{48}{20}}{\binom{52}{20}} - \frac{\binom{48}{30}}{\binom{52}{30}}$$

$$= \frac{48!}{20!28!} \cdot \frac{20!32!}{52!} - \frac{48!}{30!18!} \cdot \frac{30!22!}{52!}$$

$$= \frac{29 \cdot 30 \cdot 31 \cdot 32}{49 \cdot 50 \cdot 51 \cdot 52} - \frac{19 \cdot 20 \cdot 21 \cdot 22}{49 \cdot 50 \cdot 51 \cdot 52} \approx 10,58\%$$

6. Ако  $X_1, \dots, X_6$  са числата на заре  
 $X_1, \dots, X_6 \stackrel{\text{nd}}{\sim} U(\{1, \dots, 6\})$ . Тогава

$$P(\text{сума е 6}) = E[1_{\{\text{сума е 6}\}}]$$

$$= \mathbb{E} \left[ \mathbb{E} \left[ \frac{1}{6} \mathbb{1}_{\{X_1=6\}} + \dots + \frac{1}{6} \mathbb{1}_{\{X_6=6\}} \mid X_1, \dots, X_5 \right] \right]$$

$$= \mathbb{E} \left[ \mathbb{1}_{\{X_i=6\}} \right] = \frac{1}{6}.$$

Може и да се разпознае по #6-та, която е една от числата 1, ..., 6.

$$\mathbb{E} \left[ \text{изразът по произв} \equiv 0 \pmod{6} \right] = \infty!$$

(или може да напишете, че не съществува)

Възможно е да започнем за да кажем само  $\subset \mathbb{Z}$  -та, тогава всяко произв. от изходите може да се гледа на 6!

За 3: Coupon collector. Век упростило

$$\text{От: } 10 \left( 1 + \frac{1}{2} + \dots + \frac{1}{10} \right) \approx 29,29.$$

$$2. X_i \sim \text{Ge} \left( \frac{1}{10} \right)$$

$$X = \max \{X_1, \dots, X_{10}\}$$

$$3. \max \{x_1, \dots, x_n\} = \sum x_i - \sum \min \{x_i, x_j\} + \sum \min \{x_i, x_j, x_k\} + \dots +$$

$$+ \sum (-1)^{k+1} \min\{x_1, \dots, x_n\}$$

4.  $Ge(p_1 + p_2 - p_1 p_2)$ . Есен упрощения.

5. За 4 поколения, с вероятности

$$\frac{1}{10} \mid \frac{2}{10} \mid \frac{3}{10} \mid \frac{4}{10}$$

$\min\{X_i, X_j\} \sim Ge(p_i + p_j)$ , но това  
не следва от 4, т.к.  $X_i \not\perp X_j$ ,  
а е важно за друго изчисление.

$$\begin{aligned} \Rightarrow E[\max\{X_1, \dots, X_4\}] &= E[2X_i] \\ &= E[\sum \min\{X_i, X_j\}] \\ &\quad + \dots \end{aligned}$$

$$\begin{aligned} &= \left[10 + \frac{10}{2} + \frac{10}{3} + \frac{10}{4}\right] - \left[\frac{10}{3} + \frac{10}{4} + \frac{10}{5} + \frac{10}{5} + \frac{10}{6} + \frac{10}{7}\right] \\ &\quad + \left[\frac{10}{6} + \frac{10}{7} + \frac{10}{8} + \frac{10}{9}\right] - 1 \end{aligned}$$

$\equiv \dots$

~~2022~~ 2 1. Here  $A_i = \{\text{surfer } i \text{ never 0 waves}\}$

$$P(A_1 \cup A_2 \cup A_3) = P(\text{none 1 never 0})$$

$$= \sum P(A_i) - \sum P(A_i \cap A_j) + \sum P(A_i \cap A_j \cap A_k)$$

$$= 3P(A_1) - 3P(A_1 \cap A_2) + P(A_1 \cap A_2 \cap A_3)$$

$$= 3 \sum_{l=0}^{\infty} \left(\frac{1}{2}\right)^{l+1} \left[ \left(\frac{2}{3}\right)^l - \left(\frac{1}{3}\right)^l \right] + \frac{1}{2}$$

$$= \frac{3}{2} \cdot \left[ \frac{1}{1-\frac{1}{3}} - \frac{1}{1-\frac{1}{6}} \right] + \frac{1}{2}$$

$$= \frac{3}{2} \cdot \left[ \frac{3}{2} - \frac{6}{5} \right] + \frac{1}{2} = \frac{19}{20}$$

$$\Rightarrow \text{OEF} = 1 - P(\cup A_i) = \frac{1}{20} = \underline{\underline{5\%}}$$

ζαζ.2 Ηεκα  $X$  ε περαξδαρα ηα φεκσυραη  
ωρρα.

$$\text{ζα } k \geq 0, P(X=k) = \sum_{\ell=0}^{\infty} \binom{\ell}{k} \left(\frac{1}{2}\right)^{\ell+1} \cdot \underbrace{\binom{\ell}{k} \binom{1}{3}^k \binom{2}{3}^{\ell-k}}_{\substack{\text{οτ } \ell \text{ ε ζαηλατεκη } \\ \text{κ εσβ φεκσυραη } \\ \text{ωρρα}}}$$

$P(\text{γα ζαηλατεκη } \ell \text{ κ εσβ οτ καηηκοσ})$

$$\text{Ηεκα } \ell = k+m = \sum_{m=0}^{\infty} \binom{\ell}{k} \left(\frac{1}{2}\right)^{k+m+1} \binom{k+m}{k} \left(\frac{1}{3}\right)^k \left(\frac{2}{3}\right)^m$$

$$= \left(\frac{1}{6}\right)^k \cdot \frac{1}{2} \sum_{m=0}^{\infty} \binom{k+m}{k} \left(\frac{1}{3}\right)^m$$

$$\text{φ-ηα γαζαηλατεκη } = \left(\frac{1}{6}\right)^k \cdot \frac{1}{2} \cdot \frac{1}{\left(1-\frac{1}{3}\right)^{k+1}} =$$

$$= \frac{1}{2} \cdot \left(\frac{1}{6}\right)^k \cdot \left(\frac{3}{2}\right)^{k+1}$$

$$= \frac{3}{4} \cdot \left(\frac{1}{4}\right)^k$$

$$\Rightarrow X \sim \text{Ge}\left(\frac{1}{4}\right) \Rightarrow \phi X = \frac{3}{\left(\frac{1}{4}\right)^2} = 12.$$